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SMA 2101: Calculus I

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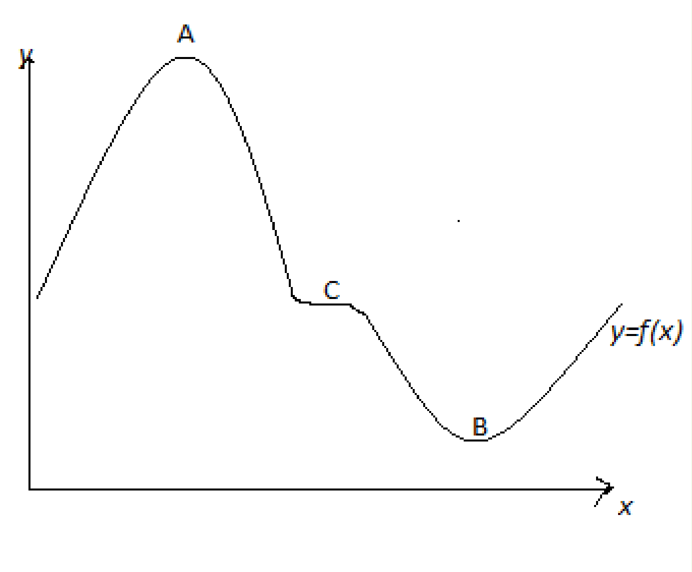
This presentation is intended to covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green Exercise tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elearning@jkuat.ac.ke.

LESSON 9

Turning Points

Consider the graph of $y = f(x)$

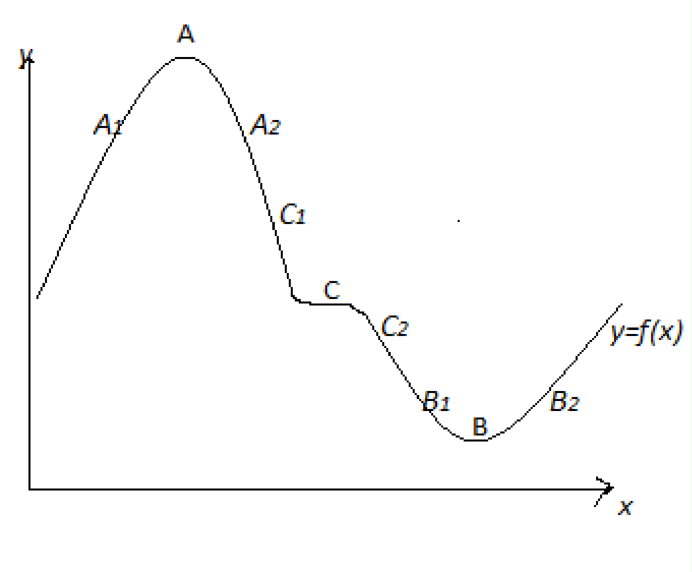


1. a maximum turning point

2. a minimum turning point

3. a point of inflection

Investigating the Nature of Stationary Values



Consider the points A_1 and A_2, B_1 and B_2, C_1 and C_2 which are left and right respectively of $A\,B\,C$ and close to them.

(1) Consider the values of y

For A (a maximum value)

y at $A_1 < y$ at A

y at $A_2 < y$ at A

For B(a minimum value)

y at $B_1 > y$ at B

y at $B_2 > y$ at B

For C (a point of inflection)

y at $C_1 < y$ at C

y at $C_1 < y$ at C

(2) Now consider the behavior of the gradient $\frac{dy}{dx}$

For A, at A_1 $\frac{dy}{dx}$ is +ve

A $\frac{dy}{dx} = 0$

A_2 $\frac{dy}{dx}$ is −ve

For B, at B_1 $\frac{dy}{dx}$ is $-$ve

B $\frac{dy}{dx} = 0$

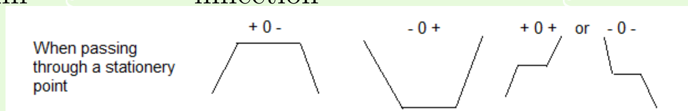
B_2 $\frac{dy}{dx}$ is $+$ve

For C, at C_1 $\frac{dy}{dx}$ is $+$ve

C $\frac{dy}{dx} = 0$

C_2 $\frac{dy}{dx}$ is $+$ve

Sign of $\frac{dy}{dx}$ maximum minimum inflection



(3) When passing through A, $\frac{dy}{dx}$ changes from positive to negative, i.e. $\frac{dy}{dx}$ decreases as x increases

$$\frac{d^2y}{dx^2} < 0 \qquad \left(\frac{d^2y}{dx^2} \text{ is negative}\right)$$

Similarly when passing through B, $\frac{dy}{dx}$ from negative to positive, i.e. $\frac{dy}{dx}$ increases as x increases

$\frac{d^2y}{dx^2} > 0$ ($\frac{d^2y}{dx^2}$ is positive)

if $\frac{d^2y}{dx^2} = 0$ we have a point of inflection

	maximum	minimum	infection
$\frac{d^2y}{dx^2}$	negative	positve	0

Example ✎. Find the points on $y = x^4 + 4x^3 - 6$ at which the gradient is zero and determine the nature of the points

Solution: $\frac{dy}{dx} = 4x^3 + 12x^2$ $\frac{d^2y}{dx^2} = 12x^2 + 24x$

At the stationery point, $\frac{dy}{dx} = 0$ therefore

$4x^3 + 12x^2 = 0$ or $4x^2(x+3) = 0$

$\Rightarrow x = -3$ or $x = 0$

When $x = -3$, $\frac{d^2y}{dx^2} = 12(-3)^2 + 24(-3) = 36 > 0$

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therefore y has a minimum value at $x = -3$

$$y = (-3)^4 + 4(-3)^3 - 6 = -33$$

Thus $(-3, -33)$ is a minimum turning point

When $x = 0$, $\frac{d^2y}{dx^2} = 12(0)^2 + 24(0) = 0$

therefore y has a an inflection at $x = 0$

$$y = (0)^4 + 4(0)^3 - 6 = -6$$

Thus $(0, -6)$ is a point of inflection

$\square$

EXERCISE 1. ✍ Find the maximum and minimum points of the curve $y = x^2(x+1) = x^3 + x^2$

Solutions to Exercises

Exercise 1. $\frac{dy}{dx} = 3x^2 + 2x \qquad \frac{d^2y}{dx^2} = 6x + 2$

At the stationery point, $\frac{dy}{dx} = 0$ therefore

$3x^2 + 2x = 0 \quad \text{or} \quad x(3x+2) = 0$

$\Rightarrow x = -\frac{2}{3} \quad \text{or} \quad x = 0$

At $x = -\frac{2}{3}$, $\frac{d^2y}{dx^2} = 6\left(-\frac{2}{3}\right) + 2 = -2 < 0 \implies$ maximum value

$$y = \left(-\frac{2}{3}\right)^3 + \left(-\frac{2}{3}\right)^2 = \frac{4}{27}$$

At $x = 0$, $\frac{d^2y}{dx^2} = 6(0) + 2 = 2 > 0 \implies$ minimum value

$$y = (0)^3 + (0)^2 = 0$$

Exercise 1